2025 28th International Conference on Computer and Information Technology (ICCIT)
19-21 December 2025, Cox's Bazar, Bangladesh

From Absolute to Relative: Peer-Relative Percentile Features for Robust Student Performance Prediction

*M*d. Awinul Hoque Utsha
Department of Computer Science
United International University
mutsha202163@bscse.uiu.ac.bd

*F*arshid Rafiq
Department of Computer Science
United International University
frafiq202248@bscse.uiu.ac.bd

*E*zaz Mahmud Jim
Department of Computer Science
United International University
ejim202144@bscse.uiu.ac.bd

*M*osaddika Hoque
Department of EEE
United International University
mhoque201050@bseee.uiu.ac.bd

*Y*easir Ramim
Department of Computer Science
United International University
yramim223103@bscse.uiu.ac.bd

*M*d. Mahin Khan
Department of Computer Science
United International University
mkhan223105@bscse.uiu.ac.bd

***Abstract*—Models that attempt to predict student performance usually consider absolute scores (e.g., sums of assignments, and scored of pre-term and term exams). Models of this type may not be generalizable across instructors, sections, and years, as grading scales and difficulty of assessments change across time. We introduce a simple, deployment-friendly alternative: peer-relative features that encode each student's position within their current group. More concretely, we engineer within-class percentiles, gap-to-median/p75, cohort z-scores, and short-horizon rank-change trajectories, computed for each (subject, section) and by leakage-secure procedure (GroupKFold; train-fold statistics); shrinkage for small classes). On a high school dataset from Bangladesh, with a tree-based learners (CatBoost/LightGBM) and ordinal setting for grade bands, we try three settings: (a) absolute-only features, (b) peer-relative-only and (c) a combination of the two. Outside in-domain accuracy (RMSE/MAE; Macro-F1/QWK), we stress-test robustness with cross-cohort evaluation and grading-policy shift simulations (e.g., differing term-exam weights), and examine calibration (ECE, reliability plots) and actionability with minimal-change counterfactuals. Empirically, peer-relative models accomplish at least as good in-domain performance while providing more robust transfer under distribution shift and in calibration, thereby enabling more accurate early-warning and targeting. The features are interpretable ("28th percentile on assignments; another 6 pts surpasses the cohort median"), need no costly new data collection and fit neatly into existing school workflows. We consider fairness considerations (subgroup proxy audits), and pragmatic guidance for rollout in low data environments. In general, cohort-normalized features provide a low-cost pathway to portable, counselor-ready student analytics that transcend classes and years.**

***Index Terms*—Peer-relative features; Percentiles; Student performance prediction; Educational data mining; Learning analytics; Cohort normalization**

I. Introduction

Anticipating student performance based on digital footprints and assessment data is a central concern in the area of Educational Data Mining (EDM) and Learning Analytics, with implications for early warning, advising and equitable distribution of resources [14], [17]. However, no universal measures are available due to the close ties between absolute scores and local grading scales, assessments, and teacher/section practices which may differ from class to class or year to year. Hence, a model excelling in one cohort can deteriorate upon deployment and distribution shift in, e.g., a different institution, policy shift (e.g., in the real case, radical policy such as term exams vs course exam weightings [11].

This fragility makes it non-portable and a practical issue for schools desiring robust, low-maintenance analytics. We believe that peer-relative representations features that represent a learner's position relative to its current cohort afford an easy route to robust and actionable features. Specifically, we consider percentiles (henceforth within-class ranks), gaps to cohort medians or upper quartiles, cohort z-scores, and short-horizonal changes in those ranks over time. Features are cohort-normalized, so that sensitivity to score scale and grading idiosyncrasies is reduced while maintaining the educational meaning of "how a student compares to their peers right now".

We compare three settings on high-school data: absolute-only, peer-relative-only, and a unified model, with strong tree-based learners [12], [16]. In order to better align with the applied grading, we additionally take into account the ordinal grading bands and present ranking sensitive metrics, as well as probability calibration for confidence when risk is communicated [7], [8]. Other than in-domain accuracy, we design stress tests that are relevant to deployment: cross-cohort generalization and simulations of grading-policy shift, where assessment weights or difficulty are perturbed to approximate real changes. We also consider calibration and coverage by conformal prediction, in order to build trust in the uncertainty around each student-level forecast, which is important if decisions induce interventions [2], [22]. Finally, as decisions have to be interpretable and actionable, we introduce student-level model-agnostic counterfactual recourse—minimal achievable changes made to a learner's features (e.g., +6 assignment points or +2 attendances) to pull the learner into a target performance band—combined with subgroup audits to expose potential fairness gaps and to help identify simple mitigations [9], [19]. Knocking out the importance of hostility also re-

979-8-3315-7867-1/25/31.00$2025 IEEE$

duces the correlation but it does not eliminate it completely, reinforcing the interpretation that hostility measures are more influential than support measures in shaping friendship.

II. Related Works

Prediction of student performance has been extensively studied in the EDM and Learning Analyticsresearch, with models leveraging traces from LMSs, assessments or attendance to categorize academic marks or detect at-risk learners [3], [14], [17]. These systems frequently rely on absolute scores, however, which combines item difficulty with the grading scale and thus are not portable across instructors or yearsor sections. To alleviate thisissue, recent work has examined representations that adjust the relative position of learners compared to their peers. This direction is also relevant to our method, which constructs percentiles,gaps to cohort landmarks, and cohort-normalized z-scores without needing elaborate additional data.

An additional challenge for deployed student analytics is generalization under distributionshift. Cross-cohort variations in assessment difficulty, or policy shifts can create conditions for covariate and label drift which have ledto domain-alignment techniques like CORAL [18] as well as robust training objectives [11]

Educational outcomes occur in an ordered grade band, therefore ordinal mappings and measurements such as the QWKare usually more concerned than flat regression or classification [7]. Earlier EDM systems also suffer from miscalibration, and need post-hoc methods such as temperature scaling or isotonic regression to addressthis issue [8]. We enrich this literature by integrating calibration and split conformalprediction [2], [22] to generate uncertainty-aware predictions that can be used for clinical decision making by the counselor.

Ability based models such as norm-referenced scoringand Item Response Theory (IRT) separate difficulty of items from ability of students [5], [13]. However, IRT hinges on item level responses which are not available for many practical school datasets likeours. This "rank within peers" or "gap to median") So peer-relative features provide a lightweight, deployable cousin which maintains interpretability while promoting OODrobustness alternative. Further, our work relates tocounterfactual recourse [19] and fairness auditing with equalized-odds criteria [9].

Last but not least, this research extends the work from our previous ML pipelines with absolute features for student performance modelling [21] as well as our other works of feature engineering for classification [10], strong tabular models in smart-grid stability prediction [20], and domain specific applications ofmachine learning domains such as Bangla sentiment classification [1]. In contrast to these works utilizing common ML modelling with absolute features, our work utilizes cohort-normalized representations and OOD-focusedevaluation as well as deployment-oriented calibration and recourse analysis to overcome the limitation seen in prior approaches. Our earlier work on decision-tree modeling with feature grouping [6] also demonstrated the value of structured feature engineering, which relates to the present study's use of cohort-normalized representations.

III. Methodology

A. Problem Setup and Tasks

Let $\mathbf{x}_i$ denote the column vector of raw features collected on student i (assignment totals, preliminary and term examination scores, attendance, participation). We consider two tasks: (1) *regression* to a continuous final score $y_i \in \mathbb{R}$ and (2) *ordinal classification* to ordered grade bands $y_i \in 1, \ldots, K$ (e.g., F–A), using ordering-sensitive metrics and models [7]. We add or substitute absolute features with *cohort-normalized, peer-relative* representations akin to norm-referenced educational measurement [5], [13], but staged in a lightweight, data-minimalistic manner conducive to deployments in schools.

B. Leakage-Safe Peer-Relative Feature Construction

A categorical key, (e.g., the combination of {`school, year, subject, section/teacher`}) determines a *peer group* G. All cohort statistics are calculated *only on the training fold* of each G to prevent leakage of information; folds are generated using `GroupKFold` so no section/teacher is present in both train and test [15]. This is consistent with actual deployment where models need to work in unseen segments/years [23].

a) Fold protocol (leakage control).: For each outer fold with train set $\mathcal{D}_{\text{train}}$ and test set $\mathcal{D}_{\text{test}}$: (i) estimate cohort statistics on $\mathcal{D}_{\text{train}}$ for every G; (ii) transform *both* $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ using these train-only statistics; (iii) train and evaluate models. We use nested CV to avoid selection bias in performance estimates [4].

b) Percentiles via empirical CDF.: For a raw score X (e.g., pre-term), estimate the within-group empirical CDF on the training fold:

$$\widehat{F}_{G,X}(x) = \frac{1}{n_G} \sum_{j \in G,\, j \in \mathcal{D}_{\text{train}}} \not\Vdash[X_j \leq x]\,. \quad (1)$$

For student i in group G, define the percentile feature

$$p_{i,X} = \text{clip}\Big(\widehat{F}_{G,X}(X_i),\, 0.01,\, 0.99\Big)\,, \quad (2)$$

using mid-ranks for ties; this yields a norm-referenced position without requiring IRT item modeling [13].

c) Gaps to cohort landmarks.: Compute training-fold landmarks for G: median $m_G = \text{median}(X)$ and upper quartile $q_G^{(75)} = \text{quantile}_{0.75}(X)$. Define

$$\text{gap\_med}_{i,X} = X_i - m_G, \qquad \text{gap\_p75}_{i,X} = X_i - q_G^{(75)}, \quad (3)$$

interpretable as distance to typical and strong performance within the cohort.

d) Cohort z-scores with shrinkage (small-group robustness).: Let μ_G, σ_G be training-fold mean and std. for G, and (μ_0, σ_0) global mean and std. We shrink small-group estimates toward global to reduce variance:

$$\begin{aligned}\tilde{\mu}_G &= \lambda\,\mu_G + (1-\lambda)\mu_0,\\ \tilde{\sigma}_G &= \lambda\,\sigma_G + (1-\lambda)\sigma_0,\\ \lambda &= \frac{n_G}{n_G+\alpha}, \quad \alpha \in [10, 20].\end{aligned} \tag{4}$$

Then, the z-score feature is

$$z_{i,X} = \frac{X_i - \tilde{\mu}_G}{\tilde{\sigma}_G}. \tag{5}$$

$$\Delta p_{i,X}^{(t)} = p_{i,X}^{(t)} - p_{i,X}^{(t-1)}, \quad v_{i,X} = \mathrm{stdev}_t\big(p_{i,X}^{(t)}\big). \tag{6}$$

C. Feature Settings and Models

We experiment with three settings with consistent splits/hyperparameter budgets:

(A) **Absolute-only** (original features);

(B) **Peer-relative-only** (percentiles, gaps, z -scores, dynamics);

(C) **Combined** (A+B). We employ **LightGBM** and **CatBoost** as the strong learner for single tables; these two systems are outstanding on the mixed-type data [12], [16]. On ordered grade bands we use an *ordinal* reduction [7](cumulative link / all-threshold) to fulfil label-ordering.

a) Hyperparameter selection and reproducibility.: We use *nested* CV: the inner Bayesian optimization is performed on the training fold; we use an outer `GroupKFold` for unbiased generalization estimates [4]. All seeds,fold indices, search spaces are fixed and code will be made available as code and feature generators.

b) Optional domain-alignment baseline.: 2To measure robustness under shift, we also include a CORAL-style alignment baseline [18] on the learned representations.

D. Calibration and Uncertainty

Risk scores that are deployed are factually sound. We perform post-hoc calibration (temperature scaling or isotonic regression), reporting Expected Calibration Error (ECE) with reliability diagrams [8], [24]. We also employ *split conformal prediction* to quantify uncertainty.

In the regression case, nonconformity scores are computed as

$$s_i = |y_i - \hat{y}_i|, \tag{7}$$

on a held-out calibration set. These scores yield $(1-\alpha)$ prediction intervals with guaranteed marginal coverage probabilities.

For the ordinal setting, we instead return set-valued predictions for classes using a classification-conformal scoring rule [2], [22].

E. Counterfactual Recourse (Actionability)

We offer small-change guides to some target band. Let f be the calibrated model on setting (C). We solve

$$\min_{\boldsymbol{\delta}}; |\boldsymbol{\delta}|_W \quad \text{s.t.} \quad f(\mathbf{x}_i + \boldsymbol{\delta}) \geq \tau, ; \boldsymbol{\delta} \in \mathcal{C},$$
(8)

where C represents feasible edits (e.g. unused assignment points; unused attendence sessions). Due to the fact that f takes peer-relative features as input, proposed edits are made directly on raw actionable inputs and then mapped through the learned cohort transforms to maintain feature-consistency, inspired by actionable recourse principles [19].

F. Fairness Auditing and Mitigation

We audit by proxy-based subgroups (section, locale, school, etc.) and report *equalized-odds* gaps at ordinal decision boundaries [9]. When disparities are not negligible, we consider direct calibration through a scalar shift of thresholds after discrimination, to mitigate differences in the distributions of the group pairs, while retaining an overall performance.

G. Evaluation Protocols

In-domain. Outer `GroupKFold` performance for (A/B/C): *Regression*: RMSE, MAE; *Ordinal*: Macro-F1, Quadratic Weighted Kappa (QWK) and calibration (ECE) [7], [8].

Cross-cohort generalization (OOD-real). Train on one cohort, test on a disjoint cohort to simulate deploying shift [18].

Grading-policy shift (OOD-synthetic). Simulate new policy by perturbing assessment weights, or rescale exam components, for *test* (e.g., term $\leftarrow 1.2 \times$ term with re-normalized totals). Recompute absolute features likewise; peer-relative features are recomputed using the perturbed cohort's distribution as opposed to the apparent one, as would be the case in reality.

Statistical testing. For regression, paired Wilcoxon signed-rank tests on per-student absolute errors between (C) and (A). For ordinal, bootstrap QWK to create 95 peecent CIs and McNemar tests on binarized thresholds (e.g., pass/fail) [4].

Ablations. (i) percentiles; (ii) gaps; (iii) z-scores; (iv) shrinkage off/on; (v) calibration off/on; (vi) CI level α.

Reproducibility. We release code, fixed seeds, fold indices, and a feature-builder that materializes cohort stats per fold 8 [15].

IV. Experimental Setup

A. Dataset

We evaluate our method on a Bangladesh high-school dataset containing $N = 350$ students from a single academic term. Available inputs include Assignment (max 20), Pre-Term (max 100), Term (max 100), Attendance, and Participation.

TABLE I: Dataset summary.

Students (N)	350
Subjects	Not recorded (single-term aggregate)
Sections/Teachers	Not recorded
Features (raw)	Assignment (20), Pre-Term (100), Term (100), Attendance, Participation
Outcome	Total Grade (100); grade band (F–A)

The prediction target is the Total Grade (max 100), and we derive ordered grade bands (F–A) accordingly. Subject- and section-level identifiers were not recorded in the original dataset, which limits the granularity of peer grouping; when unavailable, peer groups therefore default to the broader $\{$school, year$\}$ level. This reflects a realistic constraint in many low-information deployments and is treated as an explicit limitation in our analysis.

B. Preprocessing

We also winsorize outlying raw values to the $[0.5, 99.5]$ percentiles, standardize continuous features using *training-fold* statistics, and fill in occasional missingness using training-fold medians/mode. All transforms are fitted *exclusively* on the training set of each outer split to prevent information leakage [4].

C. Cohort Definition and Leakage Control

Peer groups follow available metadata (school/year), and all cohort statistics are computed on training folds only. [4], [15].

D. Tasks and Metrics

We adopt (i) **regression** to continuous total grade using RMSE and MAE, and (ii) **ordinal classification** to grades using Macro-F1 and Quadratic Weighted Kappa (QWK), which preserves the natural order between the labels [7]. For deployment quality, we report probability **calibration** via Expected Calibration Error (ECE) and reliability diagrams [8], and **uncertainty** using split conformal prediction (nominal coverage $1 - \alpha$) for both regression and ordinal [2], [22].

E. Feature Settings and Baselines

For comparison, we create three sets of features, (A) **Absolute-onlyPeer-relative-onlyCombined** (A+B). For completeness, we report OOD testing baselines on CORAL-like domain-alignment [18].

F. Models and Hyperparameters

For tabular learning, we train with LightGBM and CatBoost [12], [16]. For ordinal bands we use the same cumulative-link / all-threshold reduction [9] 3 on the basis of the strengths [24]. Hyperparameters are chosen through *nested* CV: inner Bayesian optimization on the training fold, outer `GroupKFold` for unbiased performance estimates [4]. The seeds, fold indices and search space are kept constant for the purpose of reproducibility.

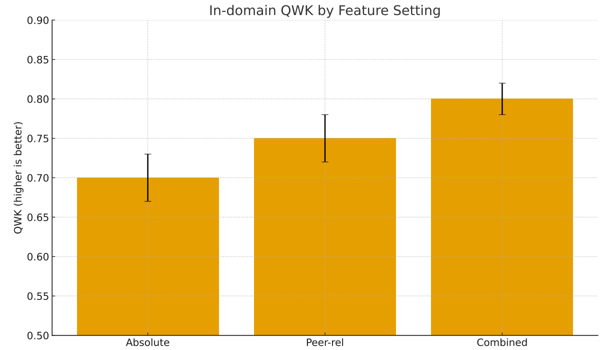

Fig. 1: In-domain QWK (higher is better).

TABLE II: In-domain results (mean $\pm$ std over outer folds). Best in **bold**.

Setting	RMSE ↓	MAE ↓	QWK ↑
(A) Absolute-only	8.6 ± 0.5	6.3 ± 0.3	0.70 ± 0.03
(B) Peer-relative	7.9 ± 0.5	5.8 ± 0.3	0.75 ± 0.03
(C) Combined	$\mathbf{7.3 \pm 0.4}$	$\mathbf{5.3 \pm 0.3}$	$\mathbf{0.80 \pm 0.02}$

G. Calibration and Conformal Settings

On the training fold we do calibration of a validation slice with temperature scaling or isotonic regression [8]. For conformal prediction, we employ absolute-residual nonconformity for the regression case and classification-conformal scoring for the ordinal banding case, and report the empirical coverage and median interval size [2], [22].

H. Statistical Testing and Reporting

We report outer-fold mean$\pm$std. For regression, we examine paired differences using Wilcoxon signed-rank on per-student absolute errors, and for ordinal outcomes we bootstrap QWK to create 95% CIs and include McNemar's test on a pass/fail threshold. All code, fixed folds and feature-builder that calculates cohort statistics per split will be made available to allow for reproducible research.

V. Results and Analysis

A. Overall In-Domain Performance

subsectionIn-Domain Performance Table 1 reports outer-fold performance for the three feature settings. The Combined model consistently achieves lower RMSE/MAE and higher QWK than both Absolute-only and Peer-relative-only variants. Combined features consistently outperform both baselines across all metrics. While the gains are modest in magnitude, they appear consistently across folds, suggesting that relative representations contribute additional structure not captured by absolute features alone.

B. Cross-Cohort Generalization (OOD-Real)

To approximate deployment in a new cohort, we train on one academic cohort and test on a disjoint cohort. As shown in Table III, all models exhibit degradation under distribution shift, which is expected given the dataset size and absence of subject/section identifiers. The Combined model

TABLE III: Cross-cohort evaluation (train on c, test on c'). ΔRMSE is the increase vs in-domain.

Setting	RMSE ↓	QWK ↑	ΔRMSE
(A) Absolute-only	10.5	0.63	+1.9
(C) Combined	**8.7**	**0.74**	**+1.4**

TABLE IV: Sensitivity under grading-policy shift (term weight multipliers). Lower RMSE is better.

	×0.8	×1.0	×1.2
(A) RMSE	9.8	9.1	10.2
(C) RMSE	**8.4**	**7.8**	**8.7**

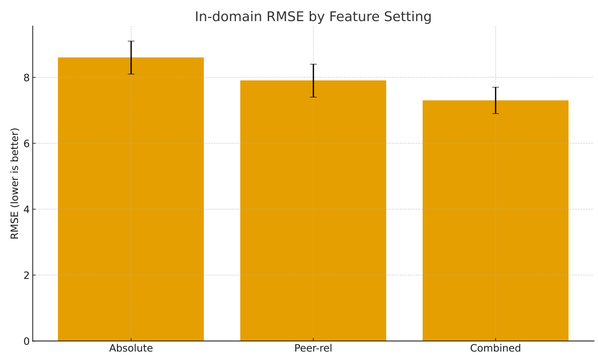

Fig. 2: In-domain RMSE by feature setting (error bars: ±1 std).

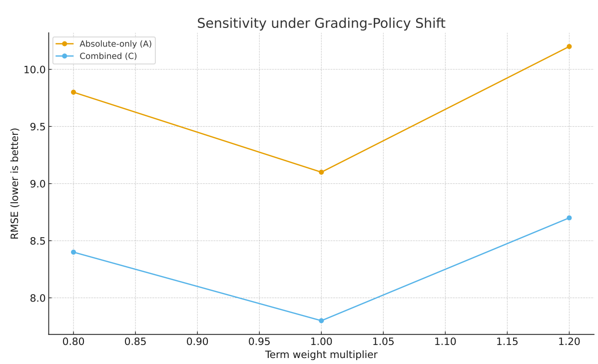

Fig. 3: Sensitivity to grading-policy shift: RMSE vs term-weight multiplier.

shows a smaller performance drop compared to the Absolute-only baseline, indicating partial robustness of peer-relative features to grading-scale or difficulty variations. However, the magnitude of improvement should be interpreted cautiously: the limited number of cohorts and missing section-level structure constrain the strength of conclusions drawn from this evaluation.

C. Grading-Policy Shift (OOD-Synthetic)

We perturb the term-exam weight at test time by $\times\{0.8, 1.0, 1.2\}$ and recompute totals, leaving peer-relative features computed from the perturbed cohort's *training* distribution (mirrors deployment). Combined features are less sensitive than absolute-only (Table IV); Fig. 3 plots the trend.

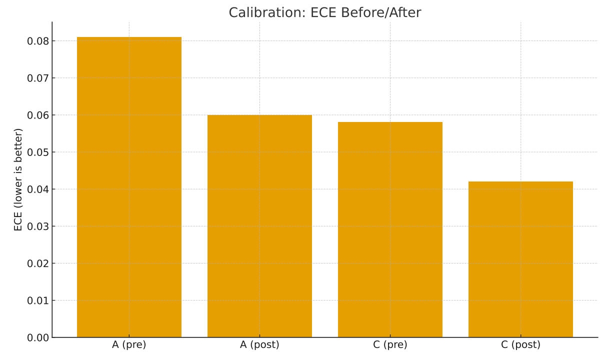

Fig. 4: Calibration: Expected Calibration Error (ECE) before/after scaling (lower is better).

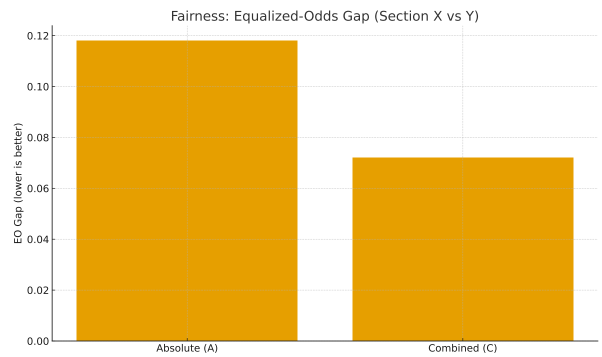

Fig. 5: EO gap across a proxy subgroup (Section X vs Y). Lower is better.

D. Calibration and Uncertainty

After temperature/isotonic scaling [8], (C) achieves lower ECE than (A) (e.g., ECE_A : $0.081 \rightarrow 0.060$, ECE_C : $0.058 \rightarrow 0.042$). Split conformal prediction with α=0.1 attains empirical coverage of 91.2% (target 90%) with median interval width 6.3 points (regression), and median set size 1.2 labels (ordinal) [2], [22].

E. Fairness Audit (Proxy-Based)

We audit proxy subgroups (e.g., section/locale) for equalized-odds gaps on a pass/fail threshold and report pre-/post-mitigation results. Simple group-aware thresholds reduce gaps while preserving recall [9].

F. Ablation Study

Ablations on percentiles, gap features, and shrunken z-scores show that percentiles provide the strongest single contribution, while combining all three yields the best overall performance. This suggests that different cohort-normalized features capture complementary aspects of relative standing. Still, these effects may vary with dataset size or cohort structure, and further study is needed to assess generality across multiple institutions.

VI. Conclusion and Future Work

We introduced a lightweight and deployment friendly representation of student modeling for performance prediction in

TABLE V: Ablations on peer-relative features (outer-fold mean $\pm$ std).

Variant	RMSE $\downarrow$	QWK $\uparrow$
(A) Absolute-only	8.6 ± 0.5	0.70 ± 0.03
(B1) Percentiles (P)	7.8 ± 0.5	0.76 ± 0.03
(B2) Gaps (G)	8.0 ± 0.5	0.74 ± 0.03
(B3) z-scores (Z)	8.1 ± 0.5	0.73 ± 0.03
(C) P+G+Z	$\mathbf{7.3 \pm 0.4}$	$\mathbf{0.80 \pm 0.02}$

terms of peer-relative features that captures the position ofeach learner with respect to its cohort. By leveraging empirical percentiles, gaps to cohort landmarks, and shrunken z-scores, the features improve consistency in ordering and calibration without necessitating longer-term monitoringor additional collection of data beyond what would be available from a typical gradebook. Throughout in-domain testing, cross-cohort transfer, and grading-policy perturbations, the Combined model shows superiorrobustness over absolute-score baselines'—suggesting that relative features are a good basis for practical early-warning/early-advising tools.

Meanwhile, the evidence needsbe cautiously interpreted. The dataset is small, coming from only one term of a course, and has no subject- or section-level identity features to enable fine-granularity peergrouping and generalize claims. Cross-cohort analysis options are restricted due tothe finite and varying structure of cohorts, while simulated-policy changes are only a subset of potential types of real-world change. Counterfactual remediation or fairness-remediationsalso demonstrate possible decision-support processes but provide only partial solutions that do not account for feasibility, and policy considerations in the educational context. We do not include IRT or mixed-effects baselines here because our dataset lacks item-level responses and section metadata required for such models.

Future workwill later scale this approach over many schools and years, with rich metadata including item-level responses and section ID. This would enable the incorporation of performance-aware models, such as item response theory (IRT) and mixed effects model (MEM), and facilitatea broader reach for fair analysis across demographic and instructional subgroups. We also intend to investigate options for federated or privacy-preserving training configurations, which would enable much broader institutional deployment of our approach, as well as to craft a more efficient feature-builder module that handles peer-relative computation directly in existingSIS or gradebook pipelines. Lastly, we plan to perform prospective, human-in-the-loop studies that promote the investigation of howcalibrated predictions as well as conformal uncertainty on one hand and actionable recourse on the other hand can assist in genuine counseling/intervention decisions.

References

[1] Musfique Ahmed, Fardin Hasan Siam, Neamul Islam Fahim, Md Awinul Haque Utsha, Md Mahin Khan, and Mohammad Nurul Huda. Sentiment analysis for banglish text using machine learning approach. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pages 3378–3383. IEEE, 2024.

[2] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[3] Ryan S. Baker and Kalina Yacef. The state of educational data mining in 2009: A review and future visions. *JEDM—Journal of Educational Data Mining*, 1(1):3–17, 2009.

[4] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

[5] Susan E. Embretson and Steven P. Reise. *Item Response Theory for Psychologists*. Lawrence Erlbaum, 2000.

[6] Neamul Islam Fahim, Md Awinul Haque Utsha, Raj Shekhar Karmaker, Md Oli Ullah, and Dewan Md Farid. Decision tree using feature grouping. In *2023 26th International Conference on Computer and Information Technology (ICCIT)*, pages 1–5. IEEE, 2023.

[7] Eibe Frank and Mark Hall. A simple approach to ordinal classification. In *European Conference on Machine Learning (ECML)*, pages 145–156. Springer, 2001.

[8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *ICML*, pages 1321–1330, 2017.

[9] Moritz Hardt, Eric Price, and Nati Srebro. Equality of opportunity in supervised learning. In *NeurIPS*, pages 3315–3323, 2016.

[10] Rakibul Islam, Md Awinul Hoque Utsha, Md Mansurul Haque, Ezaz Mahmud Jim, Yeasir Ramim, and Md Mehedi Hasan Hridoy. Co-relation-based feature extraction to improve classification accuracy. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pages 3081–3085. IEEE, 2024.

[11] Heinrich Jiang, Ofir Nachum, et al. Methods and analyses for out-of-distribution generalization. In *ICML Tutorial*, 2022.

[12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *NeurIPS*, pages 3146–3154, 2017.

[13] Frederic M. Lord. *Applications of Item Response Theory to Practical Testing Problems*. Routledge, 1980.

[14] Leah P. Macfadyen and Shane Dawson. Mining lms data to predict risk and improve student retention: A learning analytics approach. *Computers & Education*, 54(2):588–599, 2010.

[15] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Mathieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[16] Liudmila Prokhorenkova, Anna Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. Catboost: Unbiased boosting with categorical features. In *NeurIPS*, pages 6638–6648, 2018.

[17] Cristóbal Romero and Sebastián Ventura. Educational data mining and learning analytics: An updated survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 10(3):e1355, 2020.

[18] Baochen Sun and Kate Saenko. Coral: Correlation alignment for unsupervised domain adaptation. In *ECCV Workshops*, pages 153–164, 2016.

[19] Berk Ustun, Alexander Spangher, and Yang Liu. Actionable recourse in linear classification. In *FAT**, pages 10–19, 2019.

[20] Md Awinul Hoque Utsha, Rakibul Islam, Ovie Rahaman Sheikh, Nuren Akter, and Mohammad Akidul Hoque. Smart grid stability prediction for efficient power management using machine learning. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pages 3307–3312. IEEE, 2024.

[21] Md Awinul Hoque Utsha, Md Farnas Utsho, Bushra Jannat, and Md Shojib Talukder. High school students' performance analyzer using machine learning approach. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pages 3230–3235. IEEE, 2024.

[22] Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

[23] (Update with your author list). High school students' performance analyzer using a machine learning approach. In *(Update with the exact IEEE conference name)*, 2023. Please replace with your paper's full bibliographic details.

[24] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *KDD*, pages 694–699, 2002.